



MEDIAN OF TWO SORTED ARRAYS 64

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Given two sorted arrays, merge them and find the median of the combined sorted array.

Input Format

First line contains two integers n and m . Second line contains n sorted integers. Third line contains m sorted integers.

Constraints

$1 \leq n, m \leq 10^5$ $-10^6 \leq \text{array elements} \leq 10^6$

Output Format

Print the median of the merged array.

If the total number of elements is odd, print the middle element. If even, print the average of the two middle elements with one decimal place.

Sample Input 0

```
2 2
1 2
3 4
```

Sample Output 0

```
2.5
```

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Language

C++11



```
1  #include <iostream>
2  #include <vector>
3  #include <iomanip>
4  using namespace std;
5  int main() {
6      ios_base::sync_with_stdio(false);
7      cin.tie(NULL);
8      int n, m;
9      if (!(cin >> n >> m)) return 0;
10     vector<int> arr1(n);
11     for (int i = 0; i < n; ++i) {
12         cin >> arr1[i];
13     }
14     vector<int> arr2(m);
15     for (int i = 0; i < m; ++i) {
16         cin >> arr2[i];
17     }
18     vector<int> merged;
19     merged.reserve(n + m);
20     int i = 0, j = 0;
21     while (i < n && j < m) {
22         if (arr1[i] <= arr2[j]) {
23             merged.push_back(arr1[i]);
24             i++;
25         } else {
26             merged.push_back(arr2[j]);
27             j++;
28         }
29     }
```

```
29     }
30     while (i < n) {
31         merged.push_back(arr1[i]);
32         i++;
33     }
34     while (j < m) {
35         merged.push_back(arr2[j]);
36         j++;
37     }
```

Line: 46 Col: 6

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Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)

[Download](#)

1	2 2
2	1 2
3	3 4

Your Output (stdout)

1	2.5
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Expected Output

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1	2.5
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